

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MTWARA REGION



FORM FOUR MOCK EXAMINATION

CODE: 031/01

PHYSICS 01

Time: 3:00 Hours

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Year: 2026

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and two (2) questions from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. All writing must be in blue or black ink **except** drawing which must be in pencil.
5. Programmable calculators, communication devices and any unauthorized materials **are not allowed** in the examination room.
6. Write your **Examination Number** on every page of your answer booklet (s).
7. The following constants may be used,
 - i. Acceleration due to gravity $g = 10 \text{ m/s}^2$ or 10 N/kg .
 - ii. Density of water $= 1 \text{ g/cm}^3$ or 1000 kg/m^3
 - iii. Specific heat capacity of water $= 4186 \text{ J/kg}^\circ\text{C}$
 - iv. Specific heat capacity of ice $= 2144 \text{ J/kg}^\circ\text{C}$
 - v. Specific latent heat of fusion of ice $= 334\,000 \text{ J/Kg}$.
 - vi. Linear expansivity of steel $(\alpha) = 10.5 \times 10^{-6} \text{ K}^{-1}$

SECTION A (16 MARKS)

Answer all questions in this section.

1. For each of the items (i)-(ix) choose the correct answer from among the given alternative and write its letter beside the item number in the answer booklet provided.

- i. Which of the following best describes why the knowledge of physics is necessary to understand all other sciences?
 - A. Physics explains how energy passes from one object to another.
 - B. Physics explains how gravity works.
 - C. Physics explains the motion of objects that can be seen with naked eye.
 - D. Physics explains the fundamental aspects of the universe.
 - E. Physics explains how work should be done.
- ii. When a large body of experiment evidence supports or does not support a hypothesis, what may the hypothesis eventually be considered?
 - A. Observation
 - B. Insight
 - C. Conclusion
 - D. Law
 - E. Problematic
- iii. A cuboid with length 3cm, width 4cm and height 10cm is made from wood. The actual volume of the cuboid is;
 - A. 120cm^3
 - B. 240ml
 - C. 120m^3
 - D. 240cm^3
 - E. 120dm^3
- iv. Which of the following is not a force at a distance?
 - A. Electrostatic force.
 - B. Frictional force.
 - C. Force of gravity.
 - D. Magnetic force.
 - E. Nuclear force
- v. A liquid has a volume of 10cm^3 and a mass of 85g. The density of water is 1.0g/cm^3 . How does the density of the liquid compare with the density of water?
 - A. Its density is higher than that of water.
 - B. Its density is lower than that of water.
 - C. Its density is the same as that of water.
 - D. It is impossible to say with only this data.
 - E. The density is incomparable.

vi. The diagram in figure 1.0 shows the arrangement of particles in a substance.

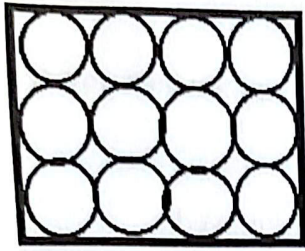


Figure 1.0

Which among the statement is the characteristic of the substance in this state?

- A. Particles move randomly and slowly and sometimes will collide against each other.
- B. The volume of an object is not fixed.
- C. Most of the particles are still in contact with one another.
- D. Difficult to be compressed.
- E. Are easily be compressed.

vii. A submarine is in equilibrium in a full submerged position, as shown in figure 2.0. What causes the up thrust on the submarine?

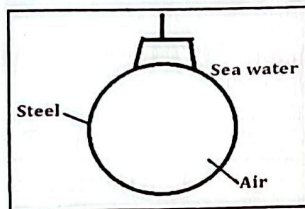


Figure 2.0

- A. The air in the submarine is less dense than sea water.
- B. The sea water exerts a greater upward force on the submarine than the weight of the steel.
- C. The submarine displaces its own volume of sea water.
- D. There is difference in water pressure acting on the top and on the bottom of the submarine.
- E. Mass of the submarine

viii. Sekela, 148cm tall stood at a distance of 2m away from a plane mirror. What will be the distance between Sekela and her image when she walks a distance of 1m away from the mirror?

- A. 2m
- B. 151m
- C. 150m
- D. 5m
- E. 6m

ix. Which one of the following shows the relationship between power and work done?

- A. Power is the ratio of work done with time.
- B. Power is the ratio of work done with distance moved.
- C. Power is the product of work done with time.
- D. Power is the ratio of time with work done.
- E. Power is the product of work done and velocity.

x. Capacitor of $1\mu\text{F}$, $4\mu\text{F}$ and $2\mu\text{F}$ are connected in series. The equivalent capacitance will be equal to:

- A. $9/4 \mu\text{F}$
- B. $4/7 \text{ F}$
- C. $7/4 \mu\text{F}$
- D. $4/7 \mu\text{F}$
- E. $9/4 \text{ F}$

2. Match the items in **LIST A** with the correspondence in **LIST B** by writing the letter of the correct response beside the item number

LIST A	LIST B
i. Movement of particles from low to high concentration. C	A. Kinetic theory of matter.
ii. Movement of particles from high concentration to low concentration F	B. Meniscus
iii. Attraction force between molecules of different substance. D	C. Convex meniscus
iv. Force of attraction between the molecules of the same substance. I	D. Brownian motion
v. It occurs as the result of adhesive force being stronger than cohesive force of attraction. J	E. Concave meniscus
vi. Tendency of matter to be in state of random motion. K	F. Osmosis
	G. Diffusion
	H. Cohesive force.
	I. Adhesive force.
	J. Elasticity.
	K. Plasticity

SECTION B (54 MARKS)

Answer all questions in this section.

3. (a) Briefly describe, how would the magnification of an astronomical telescope be affected if the focal length of the eyepiece and the objective lens are increased? (5 marks)

(b) A motor car is fitted with a convex driving side mirror of focal length 20cm. a second motor car 2m wide and 1.6m high is 6m away from the first car. Determine the position and size of the image as seen in the mirror of the first car. (4 marks)

4. (a) Explain why most vehicles have their engines directly over the drive wheel.
(b) A steel tower has a height of 324.00m at a temperature of 18°C. How tall is the tower on a day when the temperature is 35°C?

5. (a) Flask containing some warm water is connected to a vacuum pump. The pressure inside the flask is reduced. Explain why water begins to boil as the air is withdrawn (4 marks)

(b) A uniform metre rule is pivoted at its centre with a knife edge. A 20g mass is placed at the 10cm mark and a 50g mass at the 40cm mark. At what mark must a second 50g mass be placed for the system to be in equilibrium? (5 marks)

6. (a) (i) A football player kicked two balls, one light ball and one heavy ball using the same force. Which ball will accelerate more? Give reason (2 marks)
(ii) State the law or principle that supports your answer in (a) (i) (2 marks)
(b) A lever 2.4m long is to be used to raise an object weighing 500N by applying an effort of 100N as shown in figure 3.0 below. How far from the object should the fulcrum be placed? (5 marks)

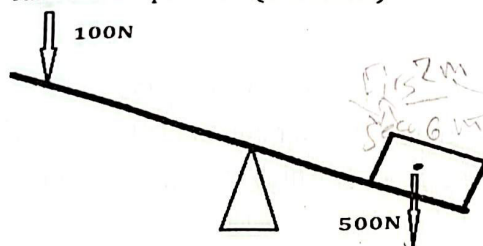


Figure 3.0

7. (a) (i) Sound of explosions taking place on the other planets is not heard by a person on the earth. explain (2 marks)
(ii) Person standing 99cm from the foot of a tall cliff, claps hands and hears an echo 0.6second later. Calculate the speed of sound in air (2 marks)
(b) Why is it not possible for heat transfer by convection to take place in solids? (5 marks)

8. (a) Explain why resistance cannot be accurately measured using just a voltmeter and an ammeter? **(4 marks)**
 (b) A house has five rooms, each with a 60W, 240V bulb. If the bulbs are switched ON from 7. 00 p.m. to 10.30 p.m., Determine the power in kilowatt-hour consumed in ten days. **(5 marks)**

SECTION C (30 MARKS)

Answer two **(2)** questions from this section.

9. (a) A cyclist moves down a steep hill without pedaling. Explain how the cyclist gain speed using the concept of energy conversion **(4 marks)**
 (b) A certain radioactive element has a half-life of 15.5 hours. If 13.125g of the element remain undecayed after 62.0hours.
 (i) How many half-life will take place? **(2 marks)**
 (ii) What was the original sample size? **(3 marks)**
 (c) With a clear labelled diagram, explain how X-rays are produced? **(6 marks)**
10. (a) Briefly explain why the fundamental frequency may alter during the day **(6 marks)**
 (b) The frequency obtained from plucked string is 400Hz when the tension is 2N. Calculate
 i. The frequency when the tension is increased to 8N
 ii. The tension needed to produce a note of frequency of 600Hz. **(4marks)**
 (c) (i) Write an expression that relates I_E , I_C , and I_B for a CE amplifier configuration.
 (ii) If, $I_C = 3mA$, and $I_E = 3.3mA$ calculate the I_B and current gain of the transistor **(5 marks)**
11. (a) Describe the mode of action of an a.c generator. **(5 marks)**
 (b) In Tanzania, electricity is distributed throughout the country by National Grid System which transmits a.c at a very high voltage. Explain what will happen if:
 (i) Direct current is to be used instead of alternating current? **(2 marks)**
 (ii) A large current is to be used instead of the high voltage? **(2 marks)**
 (c) A transformer has 10000turns in the secondary coil and 100turns in the primary coil. A current of 5.0A flows in the primary circuit when connected to a 12.0V supply.
 (i) With reason, identify the type of the transformer used. **(2 marks)**
 (ii) Calculate the voltage across the secondary coils. **(2 marks)**
 (iii) If the transformer has an efficiency of 90%, what is the current in the secondary coil? **(2 marks)**